Muhammad Ahmed

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portfolio

Experience

LLM Intern | IonIdea July 2025 - September 2025
Github: https://github.com/ahmed7077/efficient-llm-adaptation

- Fine-tuned Meta Llama 3.2–3B-Instruct on 430 instruction-response pairs using LoRA, completing 800 training steps in ~27 minutes on a Google Colab T4 GPU.
- Built end-to-end fine-tuning workflows using Hugging Face Transformers, TRL, PEFT, and Datasets, training for ~8 epochs with parameter-efficient LoRA adaptation.
- Achieved stable model convergence by reducing training loss from 1.35 to 0.51 over 800 training steps using LoRA (r=16, α=32).
- Implemented a retrieval-based routing layer to perform structured knowledge-base lookups before falling back to the fine-tuned LLM, improving domain-specific response reliability.

Projects

Facial Expression Recognition System | *Python, OpenCV, CNN, NumPy*
Github: https://github.com/ahmed7077/face-emotion-recognition

- Developed a real-time emotion recognition system using computer vision and machine learning techniques.
- Implemented face detection, feature extraction, and emotion classification using OpenCV.
- Trained and evaluated models to recognize expressions such as happiness, sadness, anger, and surprise.

Astronomical Event Detection System | *Python, Flask, OpenCV, NumPy, Raspberry Pi*
Github: https://github.com/ahmed7077/astro-event-detector-pi0w

- Built an image-based system to identify astronomical events using Raspberry Pi and Python.
- Applied image processing techniques to detect celestial phenomena from uploaded images.
- Designed a web interface to display annotated results highlighting detected events.

IoT-Based Server Monitoring System | *C++, Arduino, Sensors (DHT11, MQ2), Serial Communication*
Github:https://github.com/ahmed7077/IoT-Server-Monitoring-System

- Developed an IoT solution to monitor server room conditions using Arduino and sensor modules.
- Collected and displayed real-time environmental data for preventive maintenance.
- Enabled early detection of abnormal operating conditions through continuous monitoring.

Education

B.Tech in Information Science and Engineering
Presidency University, Bengaluru, India | CGPA:7.70 2023 – 2027

12th PUC-PCMCS
Whitefield Global School (Deeksha), Bengaluru, India | 82% 2021 – 2023

10th ICSE
Ryan International School, Bengaluru, India | 81% 2008 – 2021

Skills

Programming Languages: Java, Python, C, C++, HTML, CSS
AI/ML: Machine Learning, Large Language Models (LLMs), Natural Language Processing (NLP), Computer Vision, Convolutional Neural Networks (CNNs)
Frameworks & Libraries: Hugging Face Transformers, TRL, PEFT, OpenCV
Databases: MySQL
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems
Tools & Platforms: Git, GitHub, VS Code, Google Colab, Ollama, Raspberry Pi, Arduino
Soft Skills: Communication, Leadership, Teamwork, Adaptability

Certifications & Training

- Harvard University – CS50x: Introduction to Computer Science (In Progress)
  Completed: August 2026 | Credential: GitHub
- Generative AI Foundations | upGrad
  Completed: June 2026 | Credential: GitHub
- Java Programming & Data Structures & Algorithms Training | FacePrep – College Placement Program
  Completed: July 2026
- Python Course for Beginners: Mastering the Essentials | Scaler Topics
  Completed: April 2025 | Credential: GitHub